

Xyx Xyz

REU Proposal.docx

 Quick Submit Quick Submit Turnitin

Document Details

Submission ID

trn:oid:::1:4767935894

7 Pages

Submission Date

May 31, 2026, 10:49 AM

1,667 Words

Download Date

May 31, 2026, 10:49 AM

11,418 Characters

File Name

REU Proposal.docx

File Size

68.55 KB

24% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

-  **20 AI-generated only 24%**
Likely AI-generated text from a large-language model.
-  **0 AI-generated text that was AI-paraphrased 0%**
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer

Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?

The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.

False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?

Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.





Research Proposal

Treatment of various degrees of white spot lesions using resin infiltration—in vitro study

Student Name:

Level:

Department:

Co-supervisor:

Affiliation:

Important Point:

You should write the proposal as if you were the one who conducted the study in the article.

Introduction

Loss of minerals beneath the intact enamel surface is the first clinical sign of enamel demineralisation, appearing as opaque white areas on the surface of the teeth, known as white spot lesions (WSLs). One of the most common issues associated with fixed orthodontics is the difficulty of access for cleaning during treatment. Bacterial biofilm around orthodontic appliances can cause an increase in the production of acids, which contributes to enamel demineralization. The WSLs can be seen within a few weeks following treatment and can last for a while after the appliances are removed. In addition to being a risk factor for dental caries, these lesions have an impact on the aesthetics of the patient's smile, and therefore the patient's satisfaction, especially for anterior teeth (Patano et al., 2023; Bourouni et al., 2021).

The organic acids produced by the cariogenic bacteria begin to dissolve the calcium and phosphate ions from the surface of the enamel, and WSLs begin to form. The outer enamel may be fairly well preserved, but there are areas of porosity within the enamel, which leads to light scattering and is responsible for the characteristic chalky white appearance. The severity and visibility of WSLs are related to the depth of the lesions and the amount of mineralization lost in the lesions as compared to the mineralization of the surrounding sound enamel. This means that deeper lesions will pose a greater challenge to masking and will not be as likely to be treated conservatively. Therefore, timely diagnosis of the lesion is crucial to prevent its progression, and timely intervention is crucial before the cavitation.

There are a number of proposed management strategies for WSLs. The use of fluoride-based therapies is still popular as a way to remineralize and increase the resistance of enamel to acid attack. Similarly, minerals replenishing agents (e.g., casein phosphopeptide-amorphous calcium phosphate (CPP-ACP)) have been proven to replenish lost minerals and retard lesion progression. There are other techniques, such as enamel microabrasion, bleaching, and restoration using composite resin restorations. Not all of these, however, have their downsides. Other treatments, such as fluoride and remineralization therapies, may enhance mineralization but will not completely dissolve the esthetic appearance of lesions; other therapies, such as microabrasion and restorative therapies, may cause irreversible removal of enamel and can be more invasive (Patano et al., 2023; Marya et al., 2022).

However, all these disadvantages can be overcome by employing minimally invasive procedures for the treatment of non-cavitated WSLs, such as recently developed resin infiltration. It is a technique in which the outer enamel is hypermineralized and removed from the enamel surface, while the lesion body is porous and is infiltrated by the resin with low viscosity. The infiltrant is able to penetrate into the microporosities of the enamel and alters the refractive index of the lesion, which decreases light scattering and improves the appearance. The most popular system is the ICON resin infiltrant system that integrates hydrochloric acid etching, ethanol drying, and resin infiltration. This will not only catch the lesion, but will also provide immediate aesthetic improvement with maintenance of healthy tooth structure (Bourouni et al., 2021).

Need for the Study

Although resin infiltration of orthodontically induced WSLs has been successful in some recent studies. Resin infiltration is effective in masking lesions in systematic reviews when compared to natural remineralisation and fluoride treatments. Other findings showed that the effectiveness of treatment is related to the depth of the lesion, and there is better aesthetic improvement in superficial enamel lesions as compared with deeper dentinal lesions. Although there is increasing evidence for the effectiveness of resin infiltration, there is limited research about the optimal sequence of application of both etching and infiltrant for lesions of different depths. Major studies tend to be on the overall effectiveness of treatment and not on optimizing the protocol. Identifying the most appropriate treatment plan can provide better clinical response, prevent unnecessary erosion of enamel, and help the clinician make evidence-based recommendations for management of varying amounts of white spot lesions.

Aim of the Study

This study aims to evaluate the effectiveness of different resin infiltration methods in the color and appearance of different depths of white spot lesions (WSLs). The second objective of the study is to understand the best sequence of the different etching and infiltration with resins on each type of lesion that will result in the highest enhancement in esthetics without removing unnecessary enamel.

Null Hypothesis

The colour improvement will not be significantly different for White spot lesions of various depths after the various resin infiltration treatments.

Research Question

Are there any differences in the effectiveness of resin infiltration with lesion depth and the number of applications of etching and infiltrant?

Materials and Methodology

Ethical Approval

IREC ethical clearance will be sought prior to the study starting. All the activities of collection, storage, handling, and disposal of biological samples for research will be conducted according to the accepted ethical and biosafety procedures, as all research will be conducted using teeth extracted from humans. Teeth will only be collected after they have been extracted for orthodontic reasons, and they will be anonymized to ensure that they are kept in a confidential manner, in compliance with ethical standards (Patano et al., 2023).

Study Design

It is an experimental comparative in vitro study and will compare various resin infiltration methods in terms of colour and appearance for various depths of white spot lesions (WSLs). An in vitro design can be useful to manage experimental conditions to assess the impact of treatments, without confounding factors associated with patients.

Study Samples and Selection

Inclusion Criteria

The study sample will consist of extracted human premolars meeting the following criteria:

- Sound teeth free from caries.
- Intact buccal enamel surfaces.
- No visible cracks or fractures.
- No enamel hypoplasia.

- No previous restorations or chemical treatments.

Exclusion Criteria

The following teeth will be excluded from the study:

- Teeth affected by dental caries.
- Teeth exhibiting enamel defects or hypoplasia.
- Teeth with restorations.
- Teeth presenting visible cracks, fractures, or structural damage.

Sample Size Calculation

The study will be based on 96 human premolars that will be extracted. The lesions will be classified into four categories of teeth, with 24 teeth in each category, according to the depth of lesions. These will then be split into a set of four treatment subgroups with six teeth in each of the four. Sample size is similar to other experimental research about the success of resin infiltration, and they are enough to reveal the significance of the difference between treatments (Abbas et al., 2018).

Sample Preparation

All extracted teeth will be completely cleaned with artificial saliva, and the conditions of the mouth will be simulated. The roots will be taken off at the cemento-enamel junction with a diamond disk. The teeth will then be placed in a self-curing acrylic resin for colour evaluation, thus maintaining the same positions. Baseline colour measurements (T0) will be measured using a calibrated spectrophotometer (measured in the Commission Internationale de l'Eclairage (CIE lab*) colour system). Standardized positioning devices will be used to minimize variability in measurements and to improve the reliability (Tavares et al., 2021).

Induction and Classification of White Spot Lesions

A controlled demineralization process using an acid solution in which the mineral content of the enamel will be simulated will form white spot lesions. After induction of lesions, the depth of demineralization will then be measured using the DIAGNOdent diagnostic system. Based on the DIAGNOdent reading, the teeth will be categorized into 4 different types of lesions:

- Shallow enamel lesions
- Deep enamel lesions
- Shallow dentine lesions
- Deep dentine lesions

Based on this classification, the success of the treatment with different degrees of lesions can be determined.

Intervention Procedures

Following the categorization of lesions, each depth group will be treated with the 4 different resin infiltration treatments of the ICON resin infiltrant system, in a random manner.

Group EA

- One etching application
- One resin infiltrant application

Group EAA

- One etching application
- Two resin infiltrant applications

Group EEA

- Two etching applications
- One resin infiltrant application

Group EEAA

- Two etching applications
- Two resin infiltrant applications

All procedures will be performed according to the manufacturer's instructions, for uniformity and repeatability.

Outcome Assessment

Resin infiltration will be the primary outcome measure. Using a spectrophotometer, color measurements will be taken at three different times: baseline (T0), lesion formation (T1), and treatment (T2). The colour differences will be presented as ΔE , which can be quantitatively assessed to evaluate the aesthetic improvement. The lower the ΔE baseline to post-treatment scores, the more lesions will be masked and the better the colour recovery (Bourouni et al., 2021).

Statistical Analysis

Data analysis will be done using IBM SPSS software. All groups will have descriptive statistics (mean and standard deviation) calculated. The mean difference in the colour of the various treatment protocols will be compared to the mean difference of the various categories of lesion-depth, using One-way Analysis of Variance (ANOVA). When differences are large, then post-hoc multiple comparisons will be done to identify the differences between groups. The statistical significance of the numbers will be considered at $P < 0.05$.

Timeline for Research

Research Phase	Duration
Proposal Writing	Week 1–2
Literature Review	Week 3–4
Sample Collection	Week 5
Sample Preparation	Week 6
White Spot Lesion (WSL) Induction	Week 7–9
Resin Infiltration Procedures	Week 10–11
Data Collection	Week 12
Statistical Analysis	Week 13
Report Writing and Interpretation of Findings	Week 14–15

Research Budget

Item	Estimated Cost (OMR)
Extracted Teeth Collection	30
Artificial Saliva Preparation	20

Demineralization Chemicals	40
DIAGNOdent Usage	80
Spectrophotometer Usage	100
ICON Resin Infiltration Kits	180
Acrylic Resin Materials	25
Statistical Analysis	25
Miscellaneous Laboratory Expenses	50
Total Estimated Budget	550 OMR

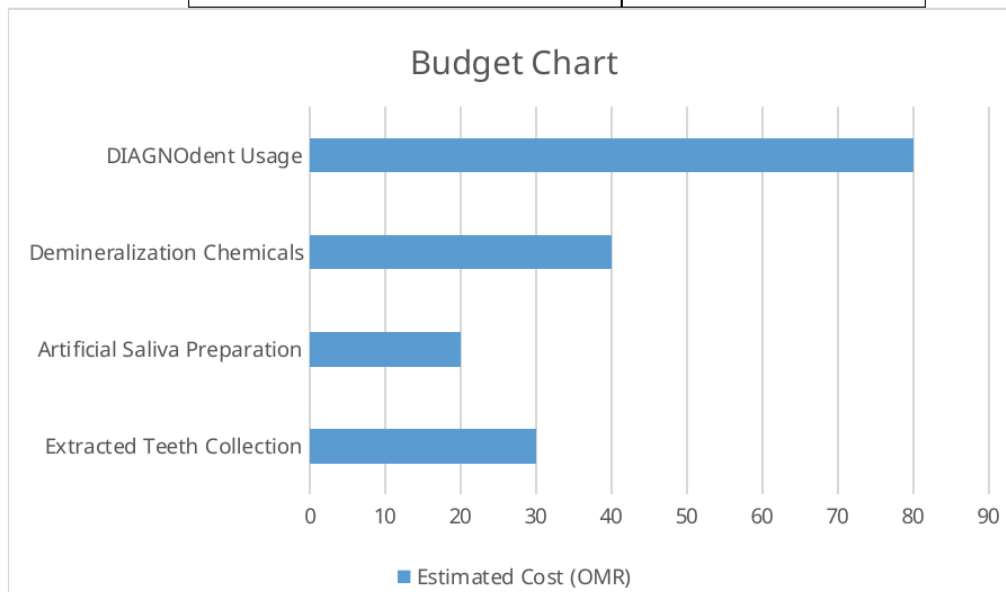


Figure 1: Budget Chart

The budget estimates are approximate and may vary depending on laboratory facilities, equipment availability, and institutional support.

Bibliography

Bourouni, S., Dritsas, K., Kloukos, D. and Wierichs, R.J. (2021). Efficacy of resin infiltration to mask post-orthodontic or non-post-orthodontic white spot lesions or fluorosis — a systematic

review and meta-analysis. *Clinical Oral Investigations*, 25(8), pp.4711–4719. doi:<https://doi.org/10.1007/s00784-021-03931-7>.

Lazar, L., Vlasa, A., Beresescu, L., Bud, A., Lazar, A.P., Matei, L., and Bud, E. (2023). White Spot Lesions (WSLs)-Post-Orthodontic Occurrence, Management and Treatment Alternatives: A Narrative Review. *Journal of Clinical Medicine*, [online] 12(5), p.1908. doi:<https://doi.org/10.3390/jcm12051908>.

Marya, A., Venugopal, A., Karobari, M.I., and Rokaya, D. (2022). White Spot Lesions: A Serious but Often Ignored Complication of Orthodontic Treatment. *The Open Dentistry Journal*, 16(1). doi:<https://doi.org/10.2174/18742106-v16-e2202230>.

Patano, A., Malcangi, G., Sardano, R., Mastrodonato, A., Garofoli, G., Mancini, A., Inchingolo, A.D., Di Venere, D., Inchingolo, F., Dipalma, G. and Inchingolo, A.M. (2023). White Spots: Prevention in Orthodontics-Systematic Review of the Literature. *International Journal of Environmental Research and Public Health*, [online] 20(8), p.5608. doi:<https://doi.org/10.3390/ijerph20085608>.

Tavares, M.I., Saraiva, J., do Vale, F., Coelho, A.S., Amaro, I.F., Marto, C.M., Francisco, I., Ferreira, M.M., Paula, A.B. and Carrilho, E.V. (2021). Resin infiltration in white spot lesions caused by orthodontic hypomineralisation: a minimally invasive therapy. *British Dental Journal*, [online] 231(7), pp.387–392. doi:<https://doi.org/10.1038/s41415-021-3476-z>.